**Code No: 117EE** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, July/August - 2023**
**LINUX PROGRAMMING**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART – A

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define the term Monolithic architecture. | [2] |
| b) | What is the need of grep command? Explain with any three options. | [3] |
| c) | What is meant by hard linking and symbolic linking? | [2] |
| d) | How the system call differs from that of the library functions? | [3] |
| e) | Mention the differences between processes and threads. | [2] |
| f) | Explain alarm() system call with brief illustration. | [3] |
| g) | What is the need of a semaphore? | [2] |
| h) | Differentiate between fifos and message queues. | [3] |
| i) | What are the applications of sockets? | [2] |
| j) | Write the similarities between UDP socket and TCP socket. | [3] |

## PART – B

**(50 Marks)**

2. Give the syntax, purpose and utilization of the following Linux commands with two options for each.
a) chmod b) uname c) du d) ps e) find. [10]

**OR**

3.a) Explain briefly about positional parameters and special variables and parameters in BASH shell.
b) Write an awk script to find the smallest of 10 integers. [5+5]

4. List and explain various directory related functions with clear syntax and example codes. [10]

**OR**

5.a) Explain fcntl( ) function with an illustrative example program.
b) Give a short notes on telldir( ) and seekdir( ) functions. [5+5]

6. Illustrate the concept of orphan processes and zombie processes in Linux with brief implementation details of system calls used for it. [10]

**OR**

7.a) Give the syntax of kill() and raise() system calls and explain.

b) What are reliable signals? Illustrate an example program for handling reliable signals. [5+5]

8. Illustrate the coprocess example by taking a simple filter to convert uppercase characters to lower case letters using popen( ) and pclose( ). [10]

**OR**

9. List and explain various APIs for system V IPC mechanism Message queues with clear syntax and example codes. [10]

10.a) Explain the implementation and use of shmctl() system call.

b) What is the purpose of shmat() and shmdt() system calls? Illustrate with an example code. [4+6]

**OR**

11. Describe the functionality of the following system calls:
a) socket( ) b) bind( ) c) accept( ) d) listen( ) e) connect( ) [10]

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